Knowledge-Grounded Reasoning and Self-Improvement for Autonomous AI Agents

Brief topic description for Program Officer discussion — ONR FY27 YIP (N0001426SF004)

Xinya Du, Assistant Professor, Computer Science, The University of Texas at Dallas xinya.du@utdallas.edu https://xinyadu.github.io

Program Officer: Dr. Behzad Kamgar-Parsi, Code 311 — Artificial Intelligence / Machine Learning, Reasoning and Intelligence

Proposed ONR S&T Focus Area: Artificial Intelligence and Autonomy — Intelligent and Collaborative Agents

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Problem. Autonomous AI agents retrieve knowledge, interpret images, call tools, and reason through multi-step plans, but can fail because one intermediate inference is wrong. For example, a port-entry assistant may read a correct tide table yet use local time as UTC when scheduling arrival. A final failure signal does not identify the mistake, and free-form self-critique may fail to determine whether the problem came from evidence, reasoning, or a tool argument. The central challenge is to teach agents to assess and improve their own reasoning while using unfamiliar knowledge.

Central hypothesis. Fine-grained, evidence-linked self-assessment can become a learning signal for autonomous reasoning: agents that know which step failed and why will learn better corrective actions and succeed more often than agents trained only on final outcomes. We will test both process-reward training and inference-time self-repair.

Research plan.

- **Ground multi-step reasoning in external knowledge.** Build agent trajectories that connect intermediate claims to retrieved passages, entity and event relations, visual observations, and tool outputs. Explicit evidence links and task constraints will expose when a step relies on missing facts or misapplies a condition, while supporting transfer to new domains.
- **Learn to diagnose reasoning errors.** Extend synthetic error injection from mathematical reasoning to agentic trajectories and multimodal tasks. Train process verifiers to distinguish hallucinated evidence, invalid inferences, entity mismatches, wrong tool arguments, and visual-grounding failures. Evaluate how well these diagnoses transfer to naturally occurring errors and unfamiliar tool interfaces.
- **Use diagnosis for self-improvement.** Train agents to use process rewards and diagnostic feedback to revise an erroneous step, continue a valid partial plan, or gather additional evidence. Study when to replan, call a tool, or abstain and how to prevent repeated self-correction from wasting computation. Compare training-time improvement against inference-only self-critique.

Evaluation and expected outcomes. MEQA, MuSiQue, and ScienceQA will probe knowledge and multimodal reasoning; controlled tool-use environments will test diagnosis and recovery. We will measure process-error localization, calibrated self-assessment, held-out task success, improvement across training rounds, and successful repair per tool call. Baselines include outcome-only training and untyped process reward models.

Naval relevance. A maritime planning agent may need to combine charts, written notices, and numerical tool outputs before making a recommendation. When it notices that a wrong time conversion invalidated an intermediate step, the agent should correct that step and continue instead of silently following a faulty plan. This addresses the Artificial Intelligence portfolio's interests in knowledge integration, uncertain reasoning, self-assessment, and planning.

Prior work and PI qualifications. Our FG-PRM (Li et al., 2025) and HiPRAG (Wu et al., 2026) develop process-level feedback for reasoning and retrieval; FaithScore (Jing et al., 2024) and FGAIF (Jing and Du, 2025) address visual-language grounding. The PI holds a 2024 NSF CAREER award and led integration of the RESIN team's high-performing DARPA KAIROS delivery system as a postdoc.